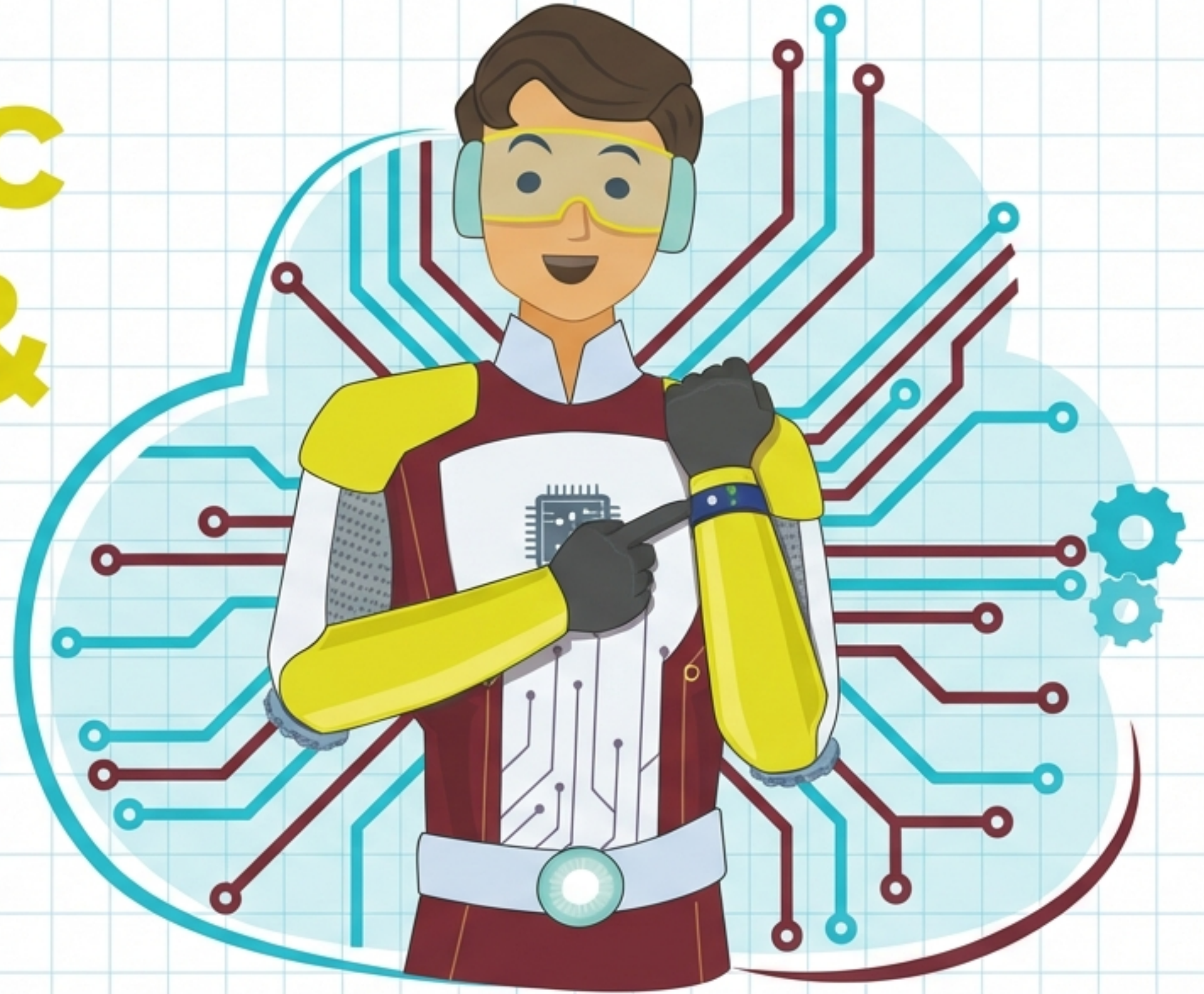
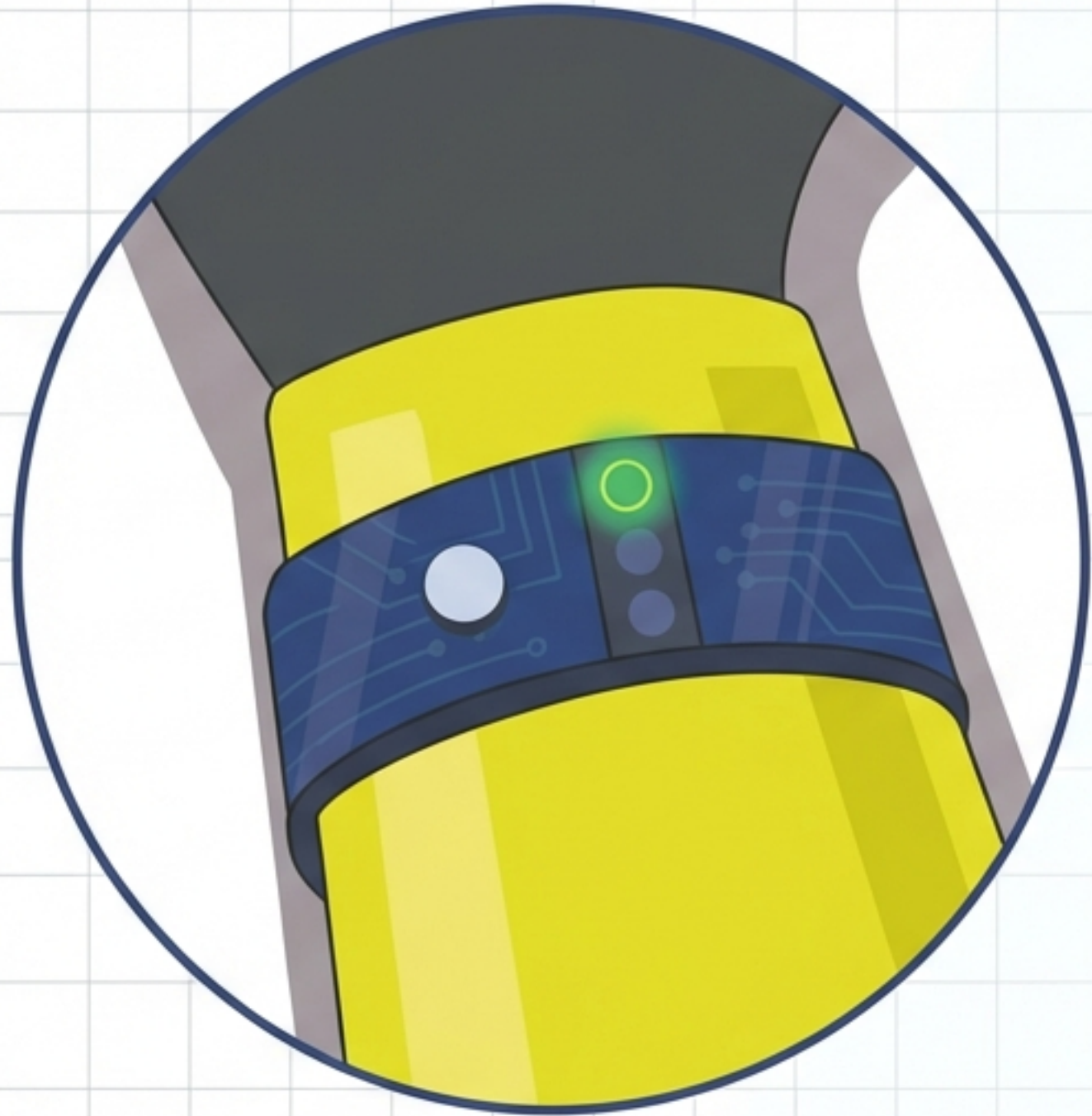


Electronic Devices & Sensors

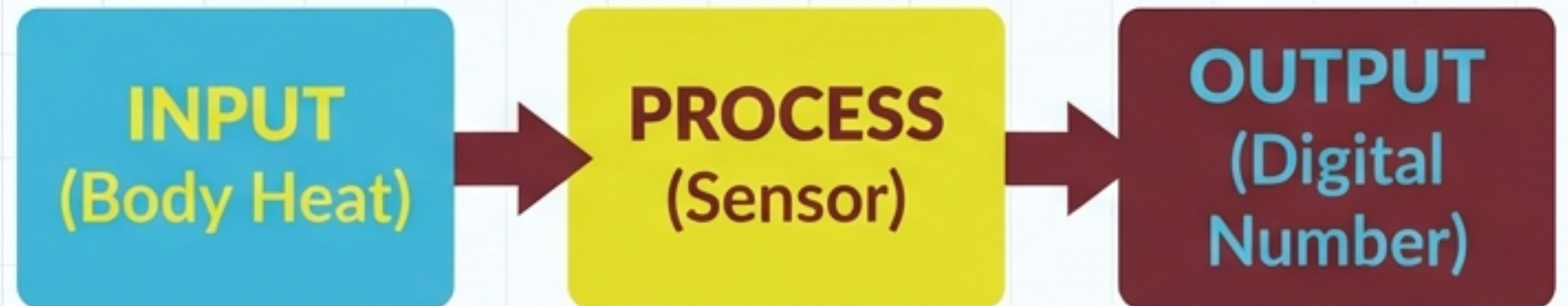
Exploring How Machines
“Feel” the World





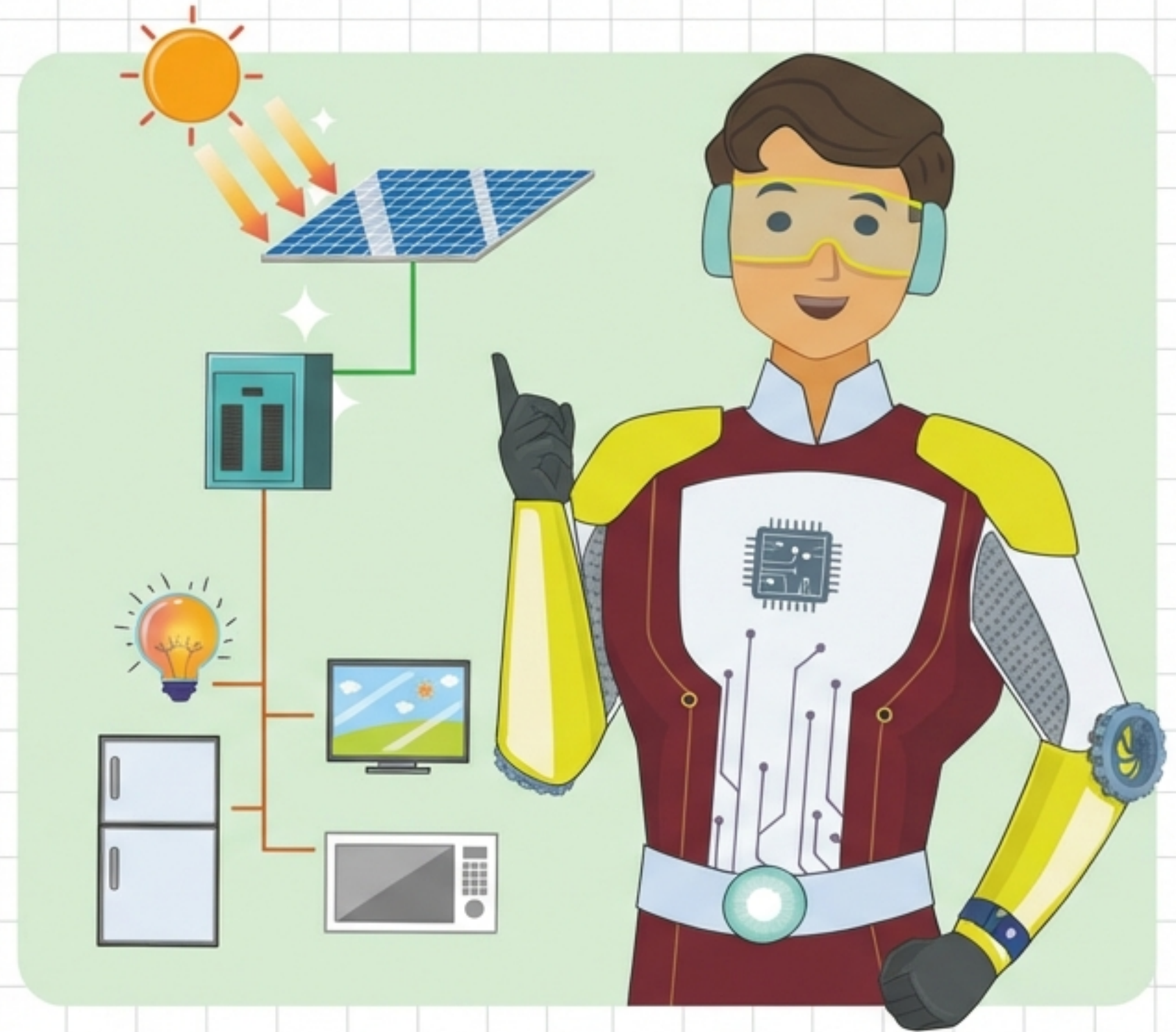
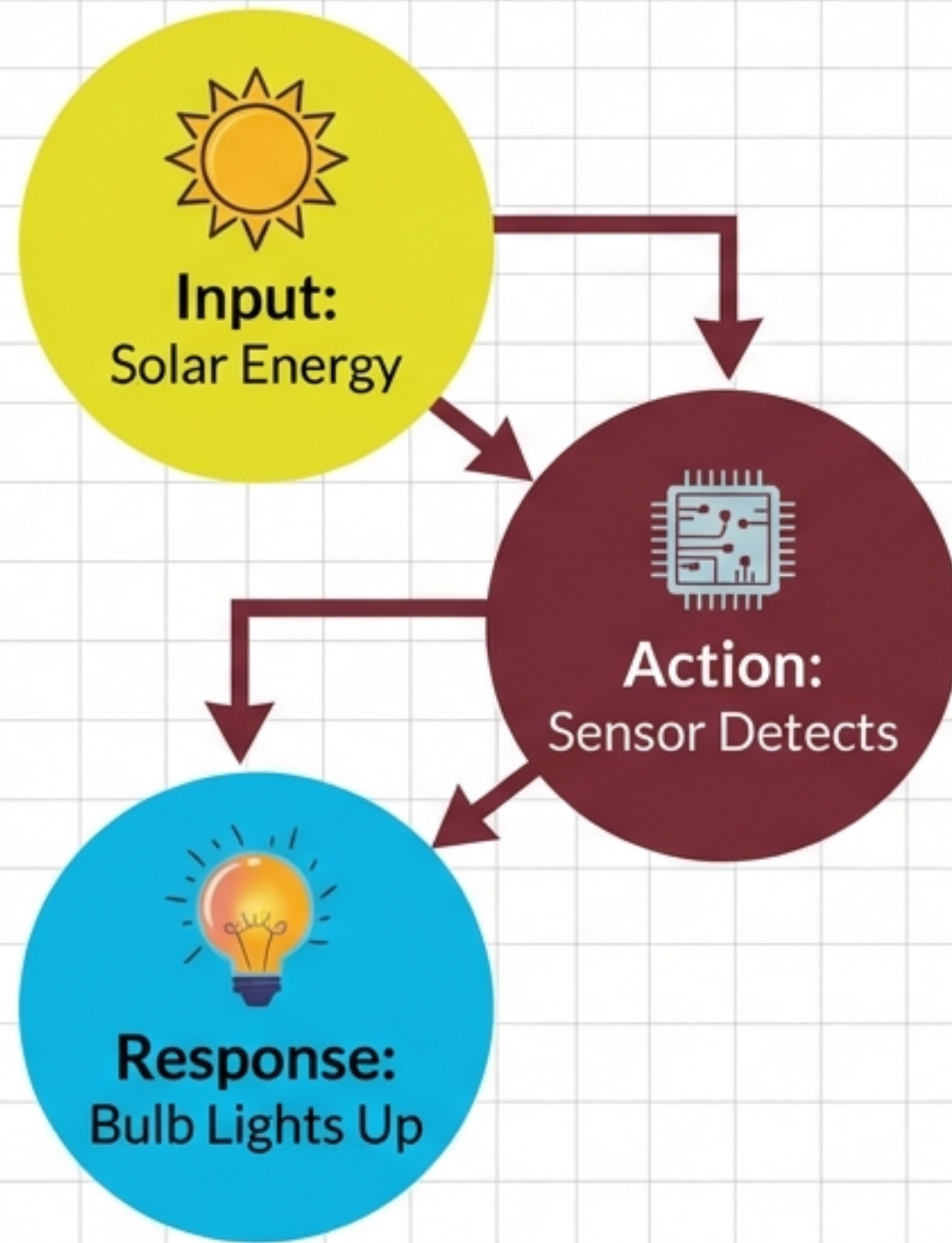
What is a Sensor?

A **sensor** is an electronic device which detects and responds to input signals from the environment.



Sensors bridge the gap between the physical world and the digital world.

How Sensors Work



Sensors convert physical signals—like light, heat, or motion—into electrical signals machines can understand.

The Vocabulary of Electronics

Transducer



Converts energy from one form to another.

Sensor



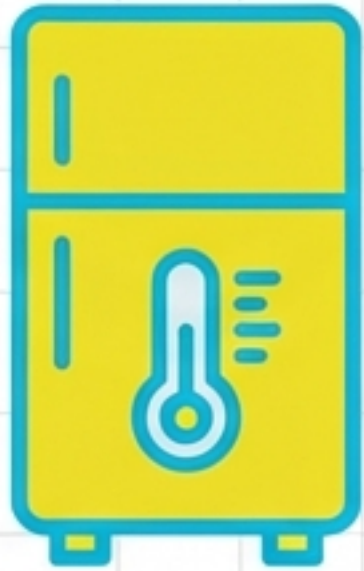
Converts physical input (light, heat, motion) into electrical energy.

Actuator



The 'Doer'. Converts electrical energy into physical output (e.g., LED, Motor).

Sensors in Daily Life: Protection & Comfort



Temperature Sensor

Maintains optimum temperature in appliances like fridges and ACs.



Smoke Sensor

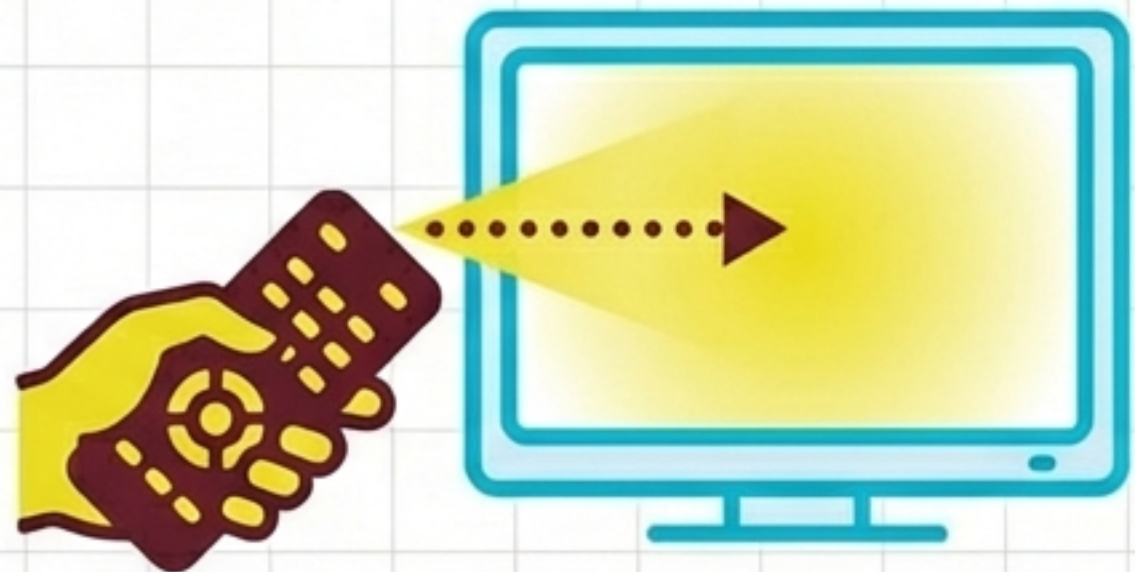
Electronic fire-protection. Automatically senses smoke in malls and apartments.

Sensors in Daily Life: Distance & Signals



Ultrasonic Sensor

Measures distance to objects using sound waves. Used for parking assistance.

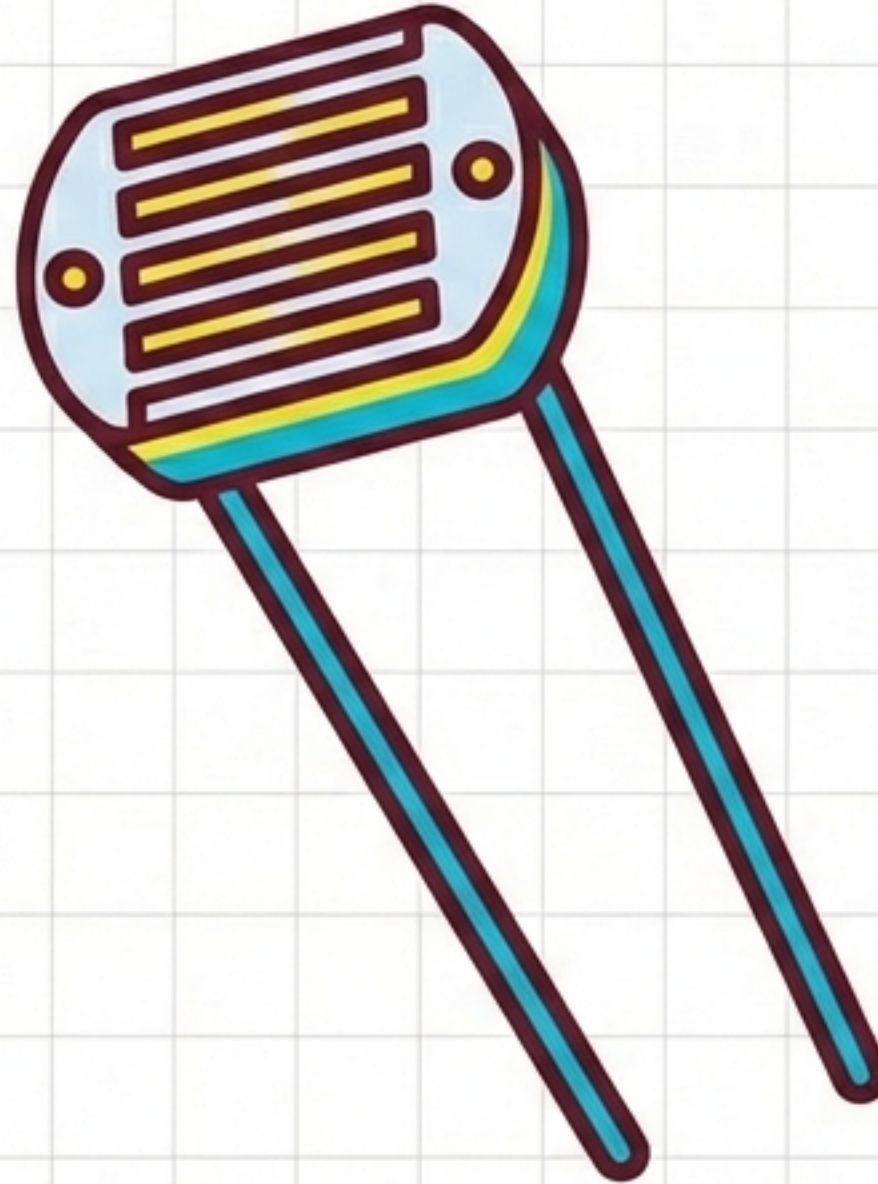


Infrared (IR) Sensor

Communicates via invisible light. Used in remotes to change channels.

Meet the LDR Sensor

Light Dependent Resistor



Bright Light =
Low Resistance
(Current Flows)

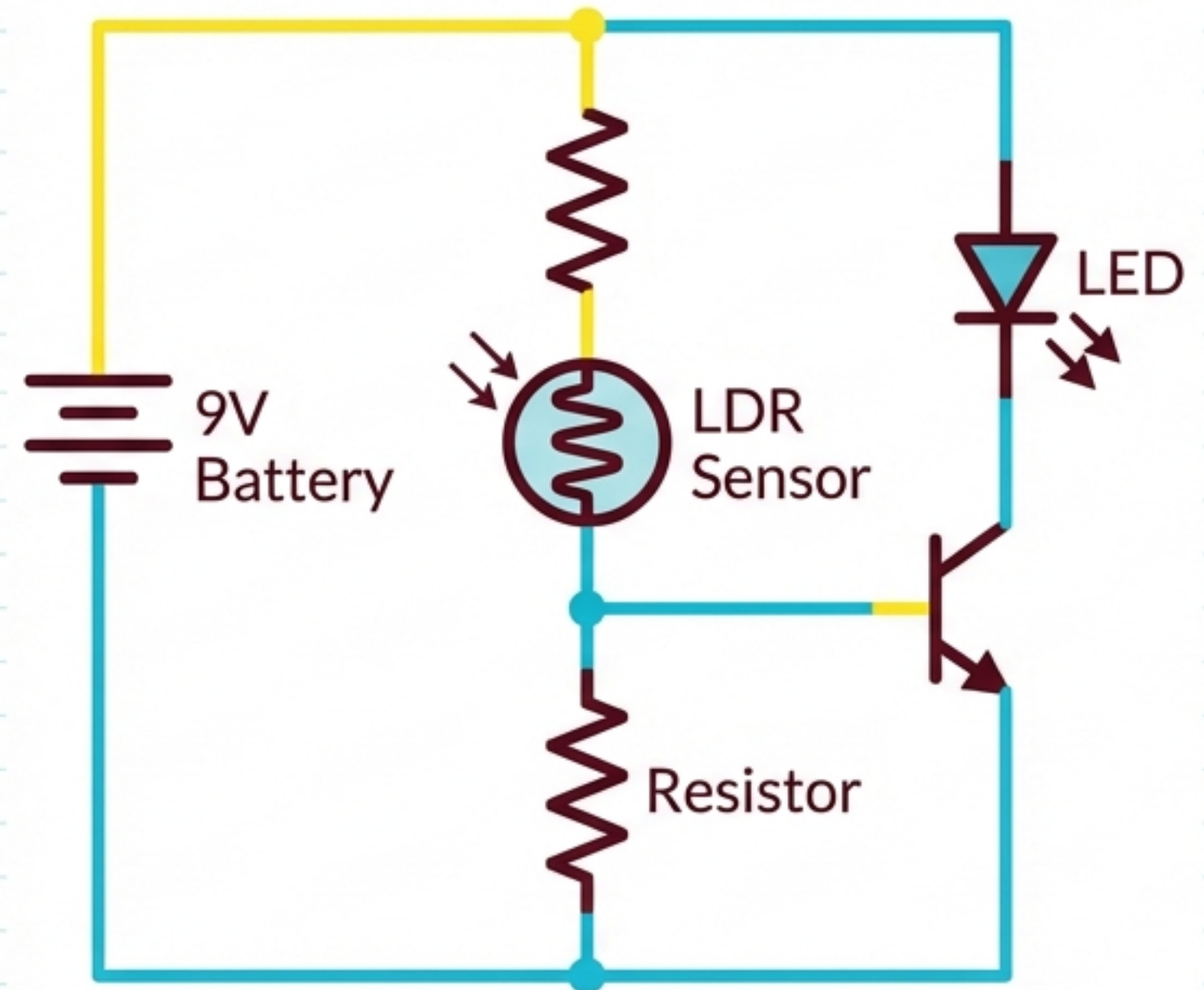
Darkness =
High Resistance
(Current Stops)

Real World Use: Used in automatic streetlights that turn on at night.

The Mission: Automatic Light Control

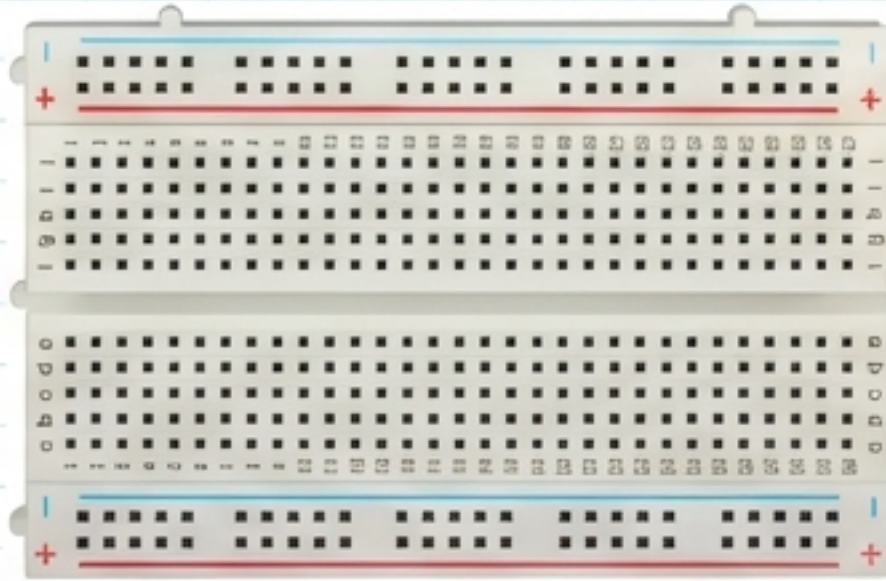
Mission Brief

- **Problem:** Leaving lights on wastes electricity.
- **Solution:** Build a circuit that “sees” the light and reacts automatically.
- **Goal:** When it is dark, the LED turns ON. When it is bright, the LED turns OFF.



Automatic light control circuit

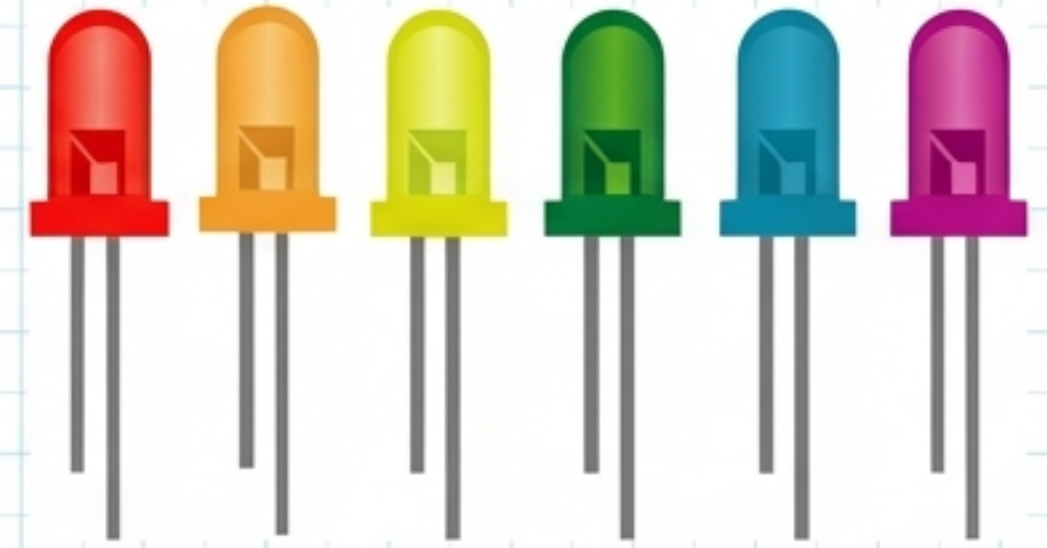
Mission Gear: Component Checklist



Breadboard (x1)



LDR Sensor (x1)



LED (x1)



Resistors
(10k Ohm & 220 Ohm)

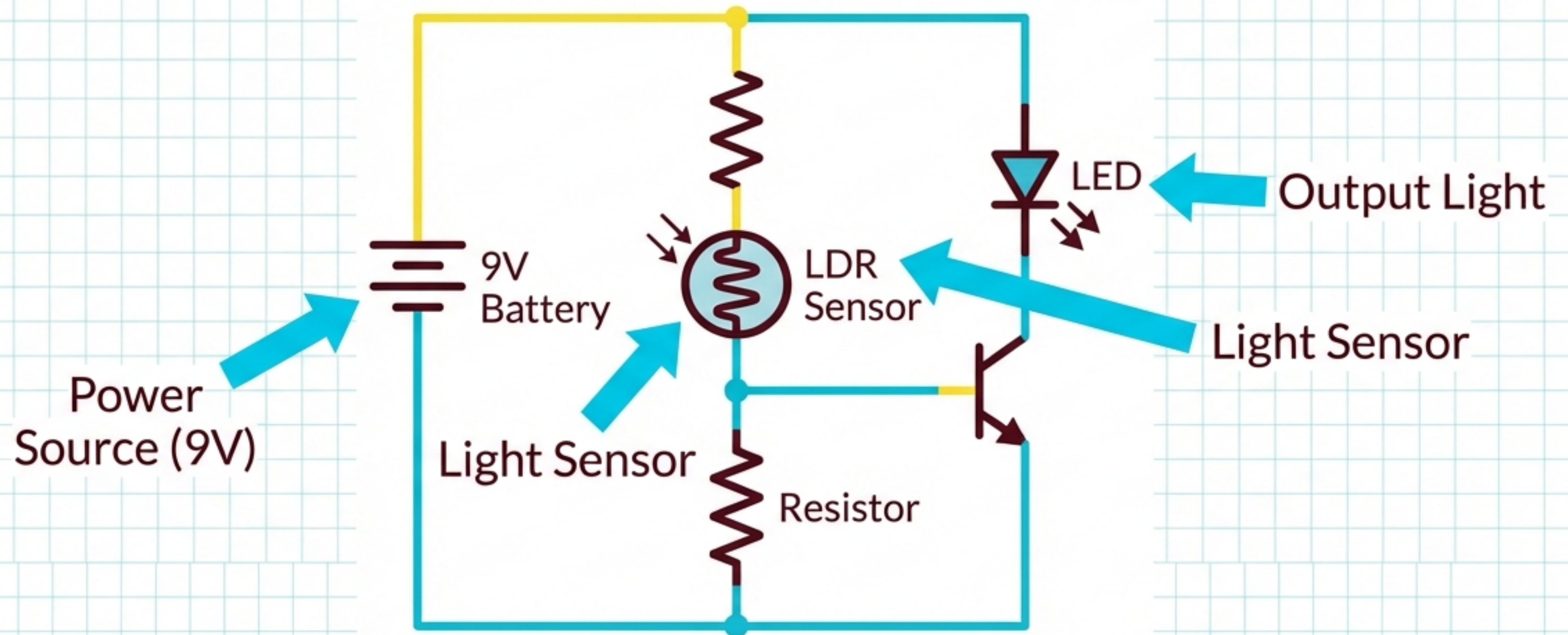


9V Battery (x1)



Jumper Wires

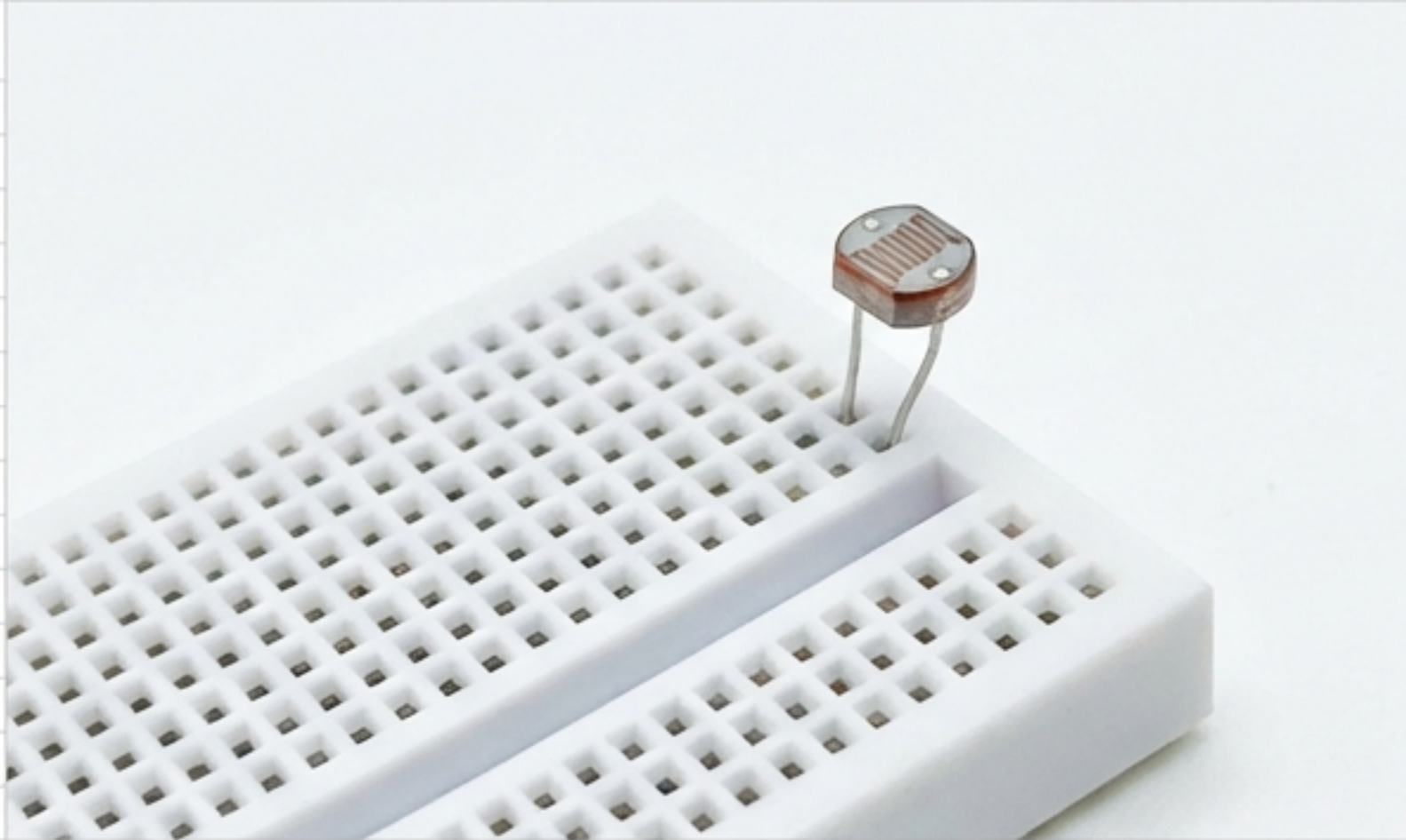
“The Blueprint”



Electricity flows from Positive (+) to Negative (-).

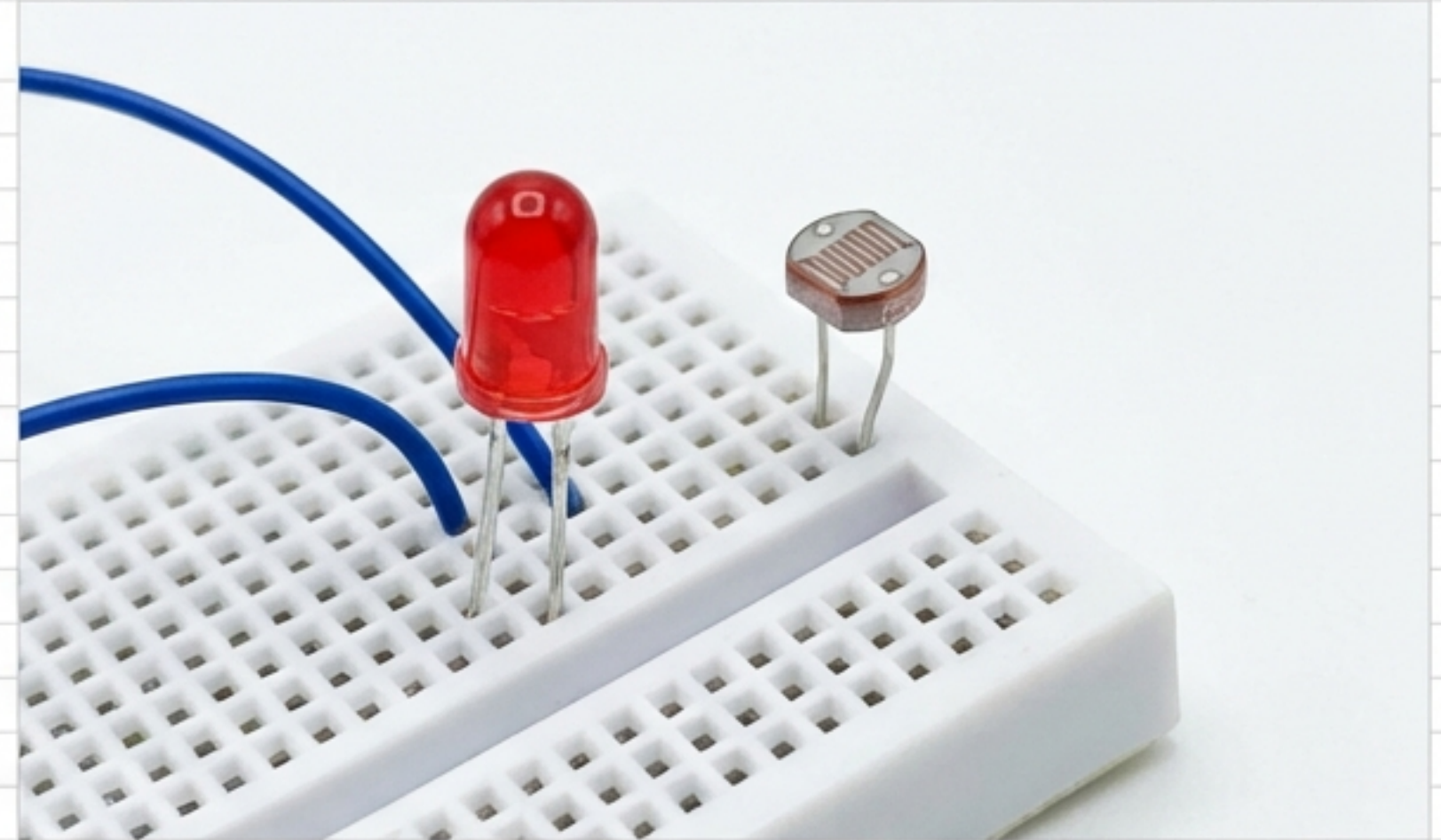
Construction: Phase 1

Step 1: Place the LDR



Take the breadboard and place the LDR SENSOR on it.

Step 2: Place the LED

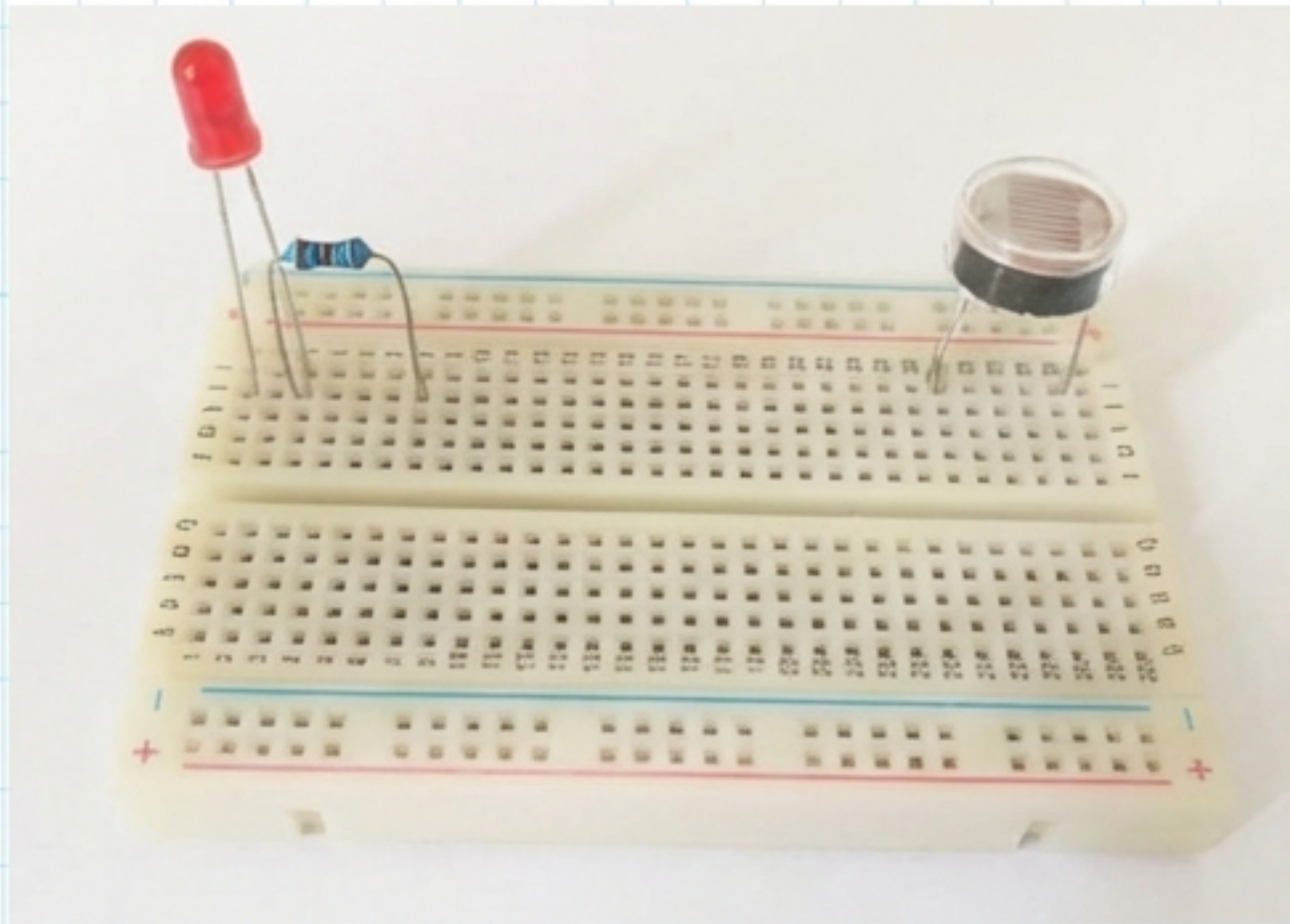


Safety Tip

Remember! The LED's longer leg is Positive (+).

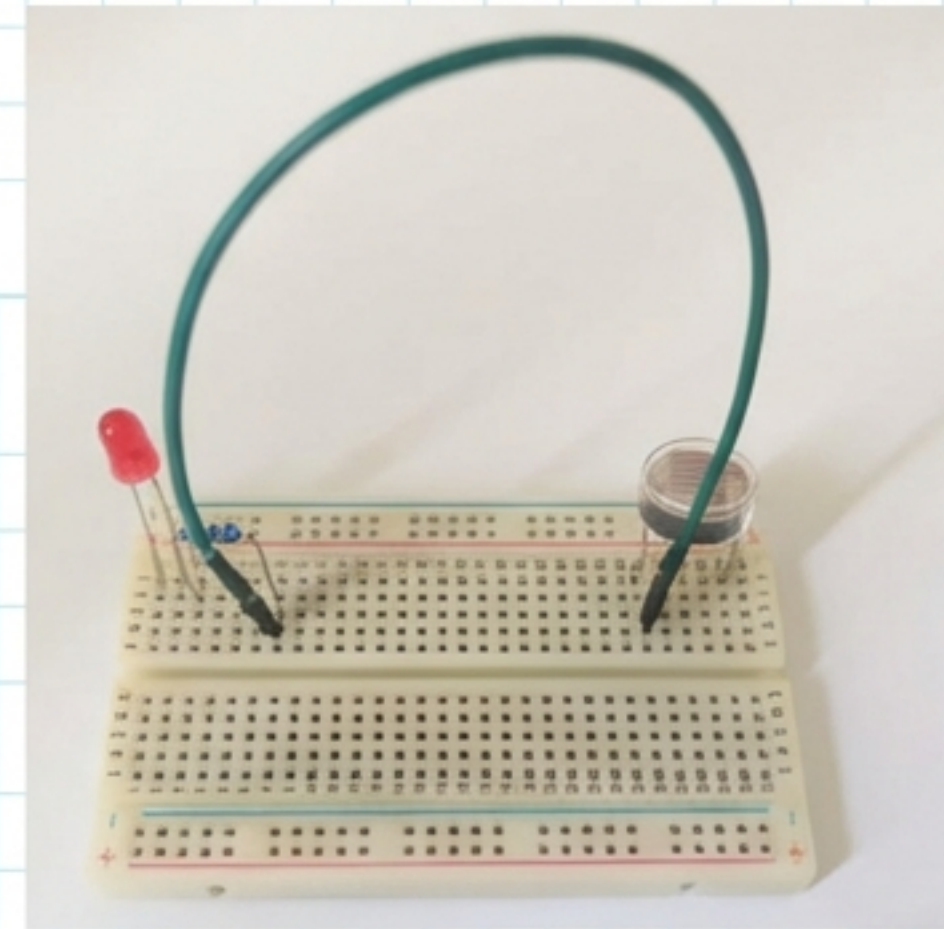
Construction: Phase 2

Step 3: Connect Resistors



Connect a 220-ohm resistor to the LED's positive leg.

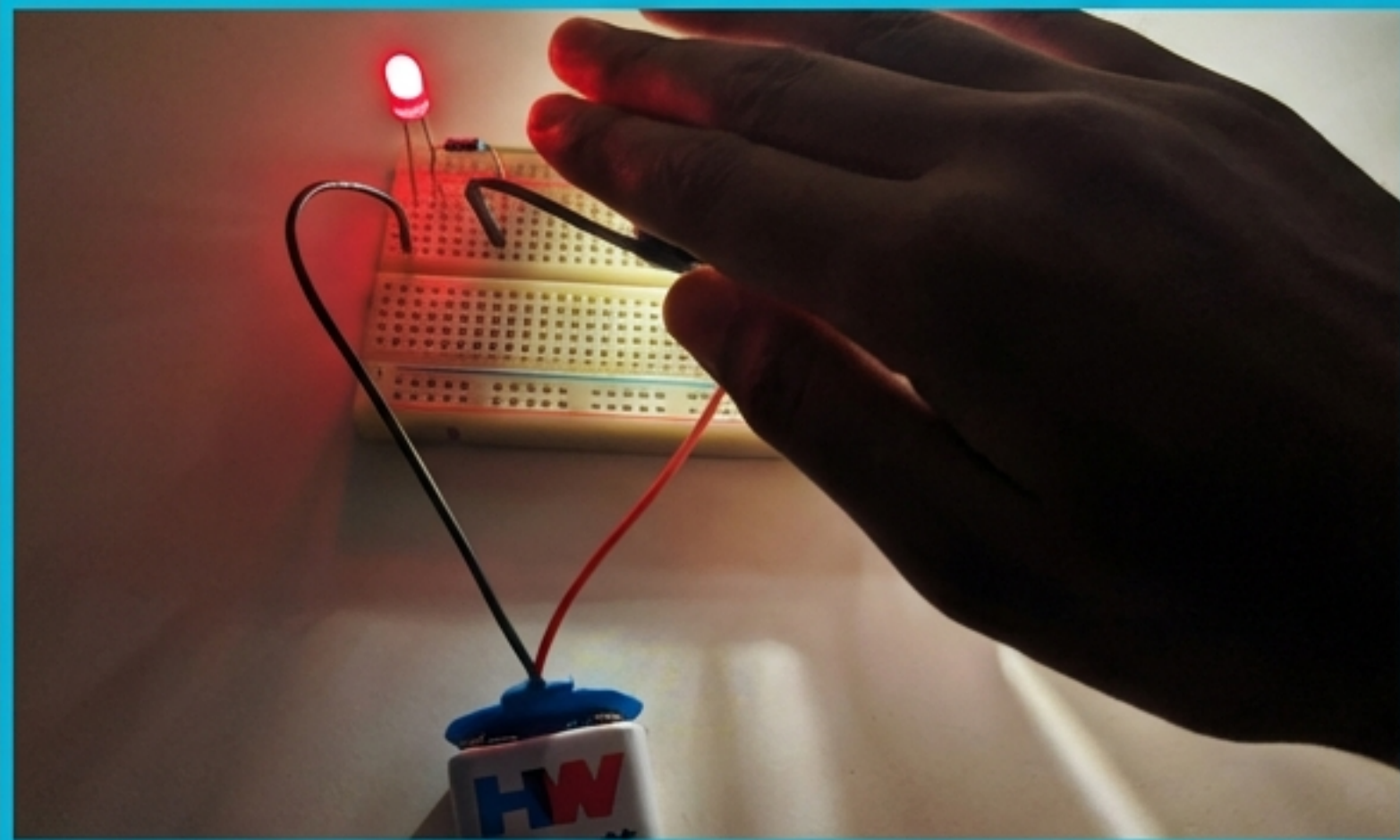
Step 4: Jumper Wires



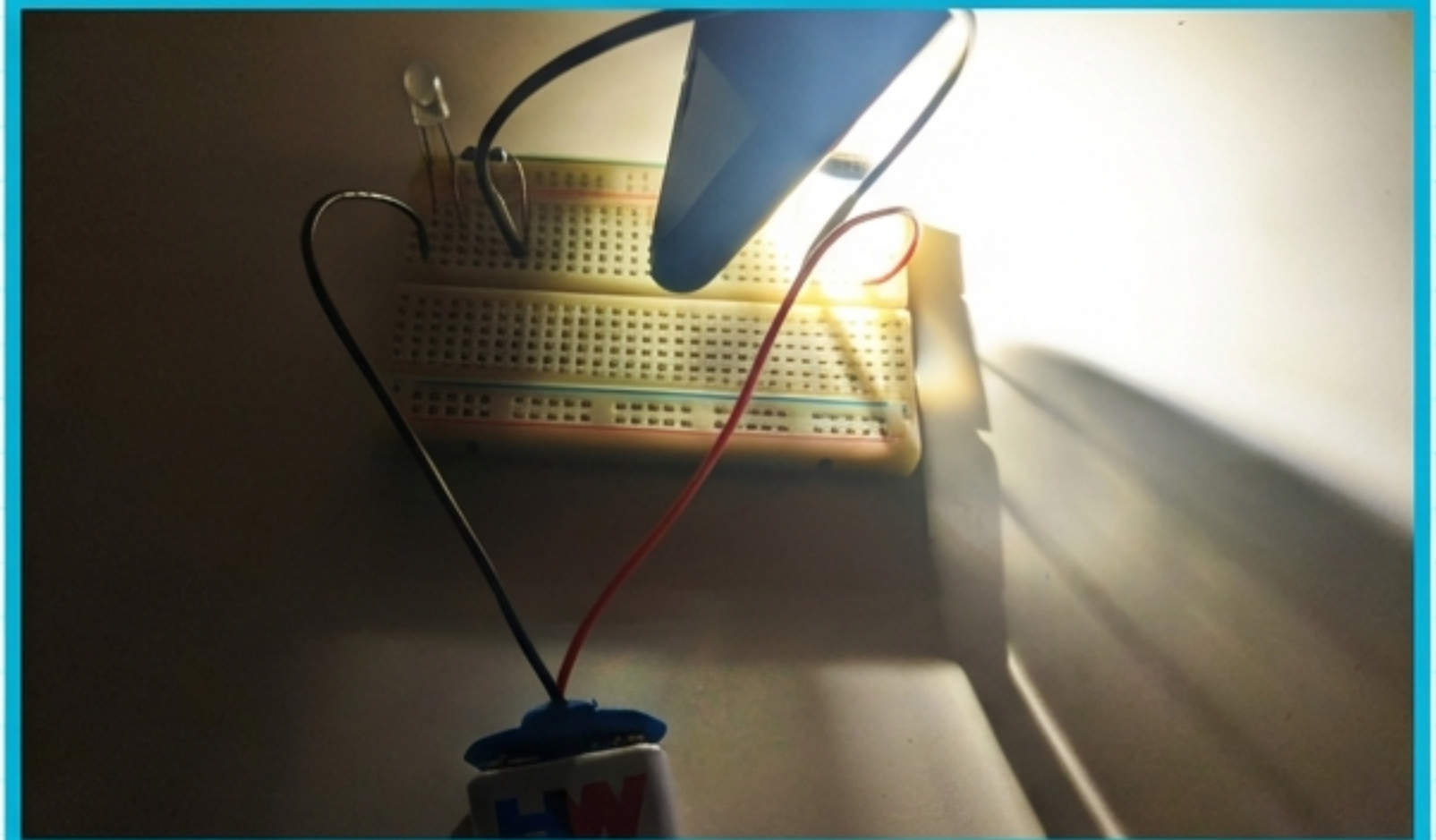
Use a jumper wire to link the resistor to the LDR.

Power Up & Test!

Scenario A: Darkness



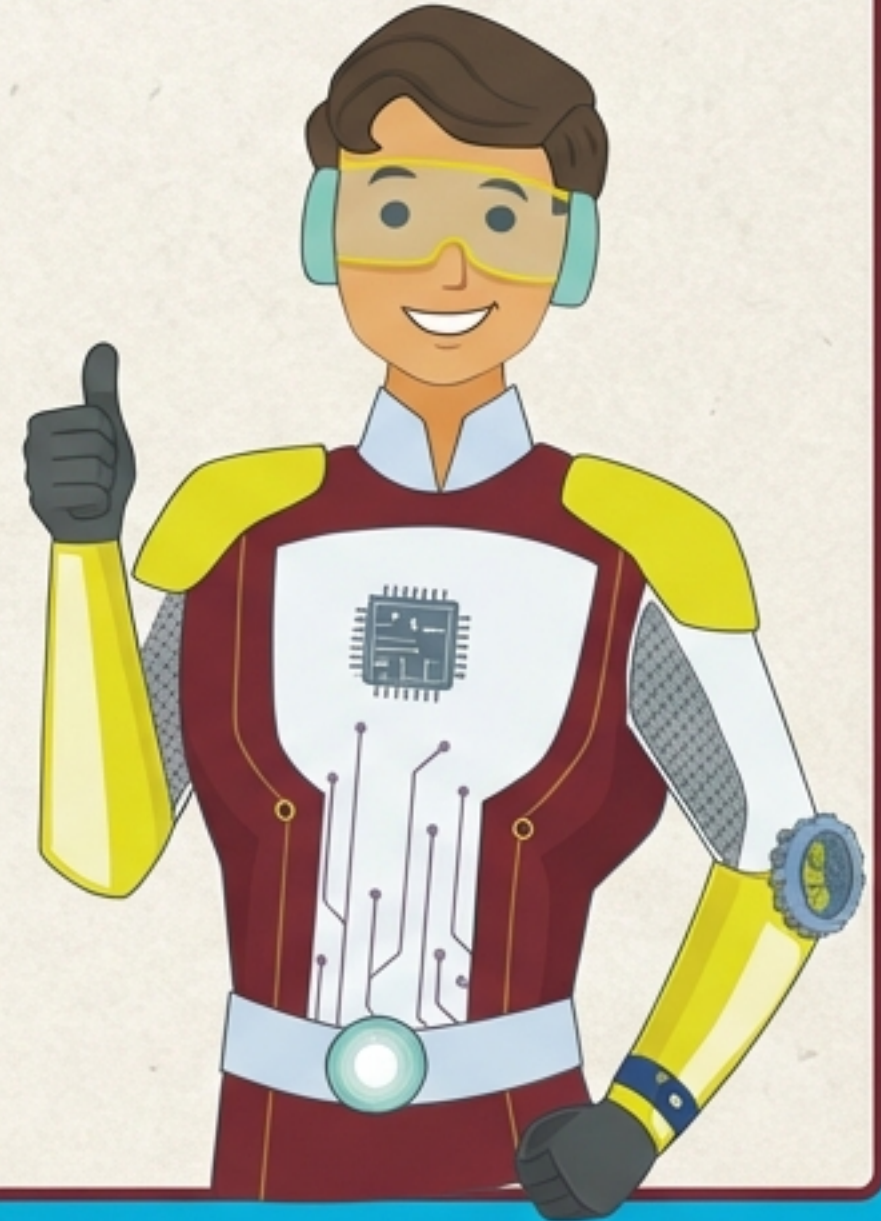
Scenario B: Bright Light



**The circuit reacts instantly
to light changes!**

Tech Flash: Key Takeaways

- **Sensor:** A device which responds to environmental input.
- **Breadboard:** A board used to hold and connect components without soldering.
- **LDR:** A light-dependent sensor that changes resistance based on light intensity.



Mission Debrief: Knowledge Check

1. **Sensors** convert environmental changes into _____ signals.
2. An **LDR sensor** indicates the **presence** or **absence** of _____.
3. If the LDR is in a **dark area**, the **LED will** turn _____.
4. The **longer leg** of an **LED** is _____.



Answers: 1. Electrical, 2. Light, 3. ON, 4. Positive